

AeroRover

Drone-Deployed Multi-Robot System for Remote Inspection and Emergency Response

Team

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GitHub

<https://github.com/LuckyKashyap001/AeroRover---Drone-Deployed-Multi-Robot-System-for-Remote-Inspection-and-Emergency-Response.git>

Rapid Deployment • Intelligent Inspection • Safer Response



The Challenge

Emergency responders and inspection teams operate in environments where direct human access is risky or impossible. Conventional approaches leave critical gaps in speed, safety, and situational awareness.

Major Challenges

Hazardous Environments

Toxic gases, fire, unstable structures, radiation, and industrial hazards endanger personnel who must physically enter to assess conditions.

Difficult Accessibility

Remote locations, damaged infrastructure, and confined spaces are hard to reach quickly with conventional equipment.

Slow Ground Deployment

Conventional ground robots require manual physical transportation to mission locations, costing critical response time.

Limited Drone Inspection

Drones provide aerial views but cannot effectively inspect ground-level or enclosed areas where hazards concentrate.

Delayed Information

Emergency teams need immediate visual, environmental, and positional data for faster, safer decisions.

Why AeroRover?

Drone

Rapid access and transportation — reaches the target zone faster than any ground-based approach.

Rover

Detailed ground exploration and inspection — enters confined, enclosed, or hazardous spaces safely.

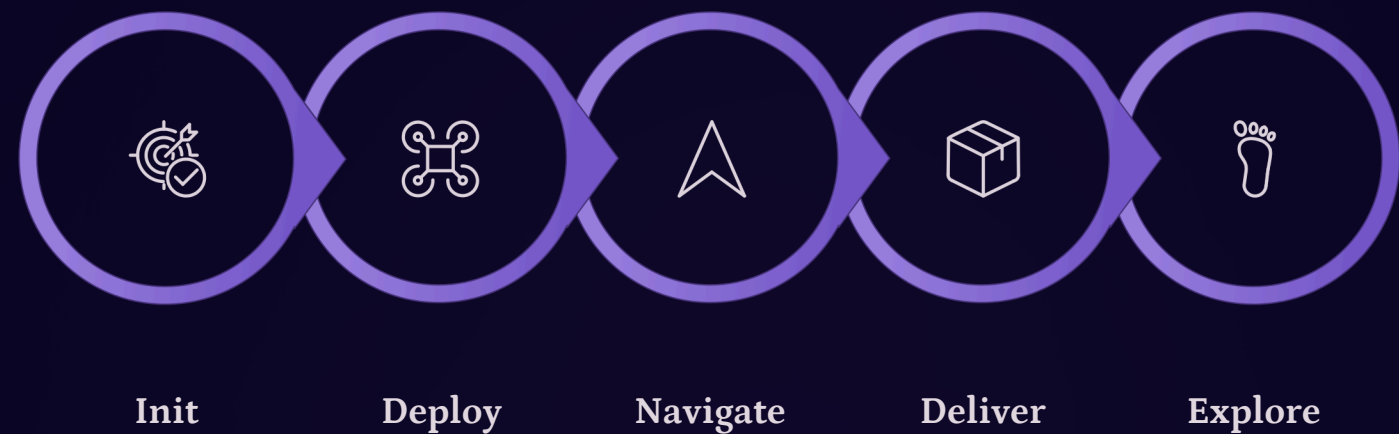
Dashboard

Real-time monitoring and control — keeps the operator fully informed at every stage.

- ☐ AeroRover bridges the gap between aerial speed and ground-level inspection depth — a capability no single-platform system can match.

How AeroRover Works

A complete end-to-end mission work flow integrating drone mobility, rover deployment, and real-time data communication into a single coordinated operation.



Each mission phase is designed to minimize operator risk while maximizing situational awareness. The drone handles rapid transit; the rover handles close-range inspection; the dashboard keeps the human operator in full control throughout.



Drone Platform: Aerial Mobility

PrimaryRole : Rapid transportation and deployment platform enabling the ground rover to reach locations faster than any conventional manual deployment method.

Main Capabilities

- Rapid access to remote or hazardous areas
- Stable flight for safe payload transportation
- Manual and autonomous flight capability
- GPS-based navigation and waypoint missions
- Autonomous takeoff and landing
- Real-time telemetry transmission
- Live camera support for aerial monitoring
- Long-range communication
- Rover payload transportation and deployment
- Return-to-home and failsafe capabilities

Key Hardware Components

STM32F411CEU6

Flight-control processing — high-performance ARM Cortex-M4 microcontroller managing all flight dynamics

1000KV Brushless Motors

Primary thrust generation — optimized for payload-carrying stability and efficient power consumption

30A ESCs

Electronic speed controllers — precise motor control for stable, responsive flight

GPS + IMU

Navigation and orientation sensing — enables waypoint missions and stable autonomous flight

Li-Po Battery + Payload Mount

Power supply and rover deployment mechanism — engineered for reliable field operation

- ❑ **Key Advantage:** Eliminates manual rover transportation to difficult locations — saving critical response time.

Prototype Overview



Note : This is just a prototype developed for proof-of-concept and system validation, focusing on verifying sensor readings, testing component integration and core functionalities, evaluating communication and control performance, assessing dimensional adaptability, and identifying potential improvements before the final system implementation.

Rover Platform: Ground-Level Inspection

Primary Role: Detailed exploration after deployment — compact size enables inspection of areas that are difficult, confined, or unsafe for human personnel.

Platform Features

- Compact and lightweight chassis for confined-space entry
- Four-wheel ground mobility for stable movement
- High-traction wheels for improved terrain handling
- Camera-based remote inspection with live video streaming
- Environmental sensor integration for hazard detection
- Low-power operation for extended mission duration
- Modular design supporting future sensor upgrades
- Manual remote operation with future autonomous capability

Processing & Camera: XIAO ESP32-S3 Sense



Camera Integration

Integrated camera support for live streaming and SD card video/image recording in the field



Edge AI Processing

On-device AI and edge-processing capability — object detection and computer vision potential without cloud dependency



Wi-Fi Connectivity

Direct Wi-Fi connection to the web dashboard — real-time data streaming to the operator interface



Low-Light Support

Low-light monitoring capability for operations in dark, enclosed, or smoke-filled environments

"Small enough to enter difficult spaces, smart enough to provide meaningful mission data."

Prototype Overview

XIAO ESP32-CAM



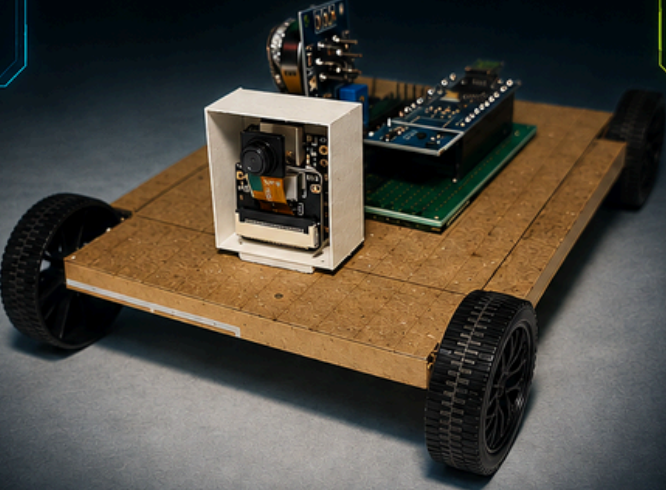
Features:

- Performs All Tasks
- Object Detection & Tracking
- Live Monitoring & Feedback on Web Dashboard
- SD Card Recording
- Low Light Recording
- Built-in Wi-Fi for Real-time Streaming & Data Transfer

ROVER

COMPACT | INTELLIGENT | RELIABLE

BUILT FOR EXPLORATION. DESIGNED FOR INSPECTION.



ROVER FEATURES

- Compact & Lightweight Design
- Small Size for Better Inspection in Congested Areas
- Durable Chassis for Rough Terrain
- Smooth & Stable Movement
- High Traction Wheels for All Surfaces
- Low Power Consumption
- Easy to Maintain & Upgrade

ENVIRONMENT MONITORING

- Temperature Monitoring
- Humidity Monitoring
- Pressure Monitoring
- Gas Detection (MQ Series)
- Real-time Data on Dashboard

AUTONOMOUS CAPABILITIES

- Autonomous Takeoff & Landing
- Waypoint Missions
- Obstacle Avoidance
- Smart Decision Making

NAVIGATION & CONTROL

- GPS-Based Navigation
- Waypoint Navigation
- Obstacle Avoidance (Optional)
- Geofencing & No-Fly Zones
- Precise & Reliable Navigation

TELEMETRY & DATA

- Real-time Telemetry Transmission
- Battery Status Monitoring
- Signal Strength Monitoring
- Distance & Movement Tracking
- Live Data on Web Dashboard

SAFETY & RELIABILITY

- Low Battery Alert & Auto Return
- Signal Loss Failsafe
- Motor Failure Protection
- Stable Movement in Various Conditions
- Redundant Systems for Safety

SIZE

Small & Compact
(Ideal for Tight Spaces)

BATTERY

Rechargeable
Li-ion Battery
(Long Runtime)

WEIGHT

Lightweight
for Easy
Mobility

CONTROL

Web / App /
Manual
Control

KEY HIGHLIGHTS

- Autonomous & Manual Control
- Live Video Streaming
- SD Card Data Logging
- Compact for Confined Space Inspection
- Efficient, Smart & Versatile Platform

APPLICATIONS

-  SURVEILLANCE & SECURITY
-  INDUSTRIAL INSPECTION
-  HAZARDOUS AREA EXPLORATION
-  ENVIRONMENT MONITORING
-  DISASTER RESPONSE & SEARCH OPERATIONS
-  INFRASTRUCTURE INSPECTION
-  RESEARCH & DEVELOPMENT

Note : This is just a prototype developed for proof-of-concept and system validation, focusing on verifying sensor readings, testing component integration and core functionalities, evaluating communication and control performance, assessing dimensional adaptability, and identifying potential improvements before the final system implementation.

Sensors, Camera & Real-Time Dashboard

Vision System

- Live video feedback for remote visual inspection of hazardous zones
- Image and video recording with SD card logging for post-mission analysis
- Low-light monitoring capability for dark or smoke-filled environments
- Object detection and tracking potential via onboard AI processing

Environmental Monitoring Sensors

Atmospheric

Temperature • Humidity • Atmospheric pressure

Chemical

Gas concentration (MQ-series) • Air-quality indicators

Operational

Distance and obstacle info • Battery voltage • Current consumption

Telemetry & Web Dashboard

Position

Drone/rover GPS coordinates and real-time movement tracking

Power

Battery level monitoring and consumption alerts for both platforms

Signal

Signal strength, connection status, and communication health

Alerts

System warnings, sensor threshold alerts, and mission status updates

Operator Monitoring Interface

Live Camera

Map / GPS

Battery

Sensors

Signal

Mission Status

Web Dashboard Preview



Note : The web dashboard preview is presented for demonstration purposes to showcase the proposed interface, live sensor monitoring, AI insights, telemetry, and system control features.

Navigation, Autonomy & Safety

Navigation Capabilities

- GPS-based position tracking and real-time location monitoring throughout the mission
- Waypoint navigation and route planning for predefined mission paths
- Geofencing and no-fly zone configuration for operational safety
- Manual override available at any point when operator judgment is required

Autonomous Capabilities

- Autonomous takeoff and landing without manual pilot input
- Predefined waypoint missions executed without continuous operator control
- Automatic return-to-home on low battery or signal loss
- Future autonomous rover navigation and obstacle avoidance
- AI-assisted mission decision-making in future development stages

Safety & Reliability Systems

Low-Battery Protection

Automatic warning and return-to-home triggered before critical power levels are reached

Signal-Loss Failsafe

Predefined failsafe behavior activates automatically on communication loss — no operator input required

Flight Stability Control

Continuous motor monitoring and flight-control processing maintains stable, predictable flight

Manual Emergency Override

Operator can assume full manual control at any moment, overriding autonomous systems instantly

Future: Redundant Safety

Additional redundant safety mechanisms planned for Stage 3 and Stage 4 development

"Autonomous when conditions are safe. Human-controlled when judgment is required."

Innovation, Advantages & Applications

Core Innovation: Three Integrated Capabilities

1. Aerial Mobility Rapidly reaches remote locations — eliminating the time cost of manual rover transportation to hazardous zones		2. Ground-Level Exploration Performs close inspection where drones are less effective — confined spaces, enclosed structures, ground-level hazards		3. Real-Time Intelligence Provides live video, sensor readings, telemetry, and mission information — keeping operators fully informed
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Key Advantages

- Faster deployment than manually transporting a rover to the scene
- Reduces human exposure to hazardous environments significantly
- Combines aerial and ground perspectives in one integrated system
- Modular architecture supports additional sensors and payloads
- Supports real-time remote monitoring and decision-making
- Expandable toward autonomous multi-robot missions
- Suitable for both inspection and emergency-response operations

Application Domains

 Search & Rescue Disaster zones and missing-person operations	 Industrial Inspection Confined spaces, pipelines, and plant facilities
 Environmental Monitoring Hazardous area exploration and air quality	 Infrastructure Bridge, tunnel, and structural monitoring
 Surveillance Security operations and perimeter monitoring	 Research Robotics development and academic study

Future Development Roadmap

AeroRover follows a structured four-stage development path — from validated prototype to a fully autonomous, scalable multi-robot ecosystem.



Stage 1 — Current Prototype

Basic drone and rover platforms • Camera integration • Sensor testing • Telemetry monitoring • Web dashboard development • Manual control and prototype validation



Stage 3 — AI & Advanced Sensing

AI-based object detection • Human/victim detection • Hazard identification • Computer vision • LiDAR or depth-camera integration • Intelligent route planning • Edge AI processing



Stage 2 — Improved Navigation

High-accuracy GPS • Advanced waypoint missions • Better position estimation • Autonomous rover movement • Improved obstacle sensing • Automatic rover deployment mechanism



Stage 4 — Fully Integrated Multi-Robot System

Multiple rover deployment • Drone-to-rover coordination • Autonomous docking and recovery • Mesh communication • Swarm coordination • Wireless charging • Cloud mission analytics • Digital twin simulation

- ❑ **Future Vision:** Develop AeroRover into an intelligent, scalable multi-robot ecosystem capable of autonomously reaching hazardous areas and supporting faster emergency decisions.

Conclusion & Thank You

AeroRover: Combining Aerial Mobility and Ground Robotics

AeroRover demonstrates how the combination of aerial mobility and ground robotics creates a more flexible, safer, and faster approach to remote inspection and emergency response — reducing human risk while improving situational awareness.

Rapid Access

Drone provides rapid access and rover deployment to locations unreachable by conventional means

Ground Inspection

Compact rover enables close-range ground inspection in confined, hazardous, or enclosed spaces

Situational Awareness

Cameras and sensors provide real-time situational awareness — telemetry and dashboard keep operator fully informed

Scalable Future

Modular architecture supports future AI, additional sensors, and full multi-robot expansion across all four development stages

"Reach Faster. Inspect Smarter. Respond Safer."

Team CyberTron — Lucky Kashyap • KIET Group of Institutions

THANK YOU